

# Functional Dependencies and Normalization

## Questions with Solutions

### Question 1

title

Consider the following set  $F$  of functional dependencies on the relation schema

$$R = (A, B, C, D, E, G) :$$

$$F = \{A \rightarrow BCD, \quad BC \rightarrow DE, \quad B \rightarrow D, \quad D \rightarrow A\}.$$

- Compute the attribute closure  $B^+$ .
- Prove (using Armstrong's axioms) that  $AG$  is a superkey.
- Compute a canonical cover for this set of functional dependencies  $F$ ; give each step of your derivation with explanations.
- Give a 3NF decomposition of the given schema based on a canonical cover.
- Give a BCNF decomposition of the given schema using the original set  $F$  of functional dependencies.

title

- a. Compute  $B^+$ :**

Start with:

$$B^+ = \{B\}$$

Use FDs to add attributes:

- From  $B \rightarrow D$ , add  $D$ :

$$B^+ = \{B, D\}$$

- From  $D \rightarrow A$ , add  $A$ :

$$B^+ = \{A, B, D\}$$

- From  $A \rightarrow BCD$ , add  $B, C, D$  (already have  $B, D$ , add  $C$ ):

$$B^+ = \{A, B, C, D\}$$

- From  $BC \rightarrow DE$ , since  $B, C \in B^+$ , add  $D, E$ :

$$B^+ = \{A, B, C, D, E\}$$

No FD involves  $G$ , so it is not added.

**Final closure:**

$$B^+ = \{A, B, C, D, E\}$$

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**b. Prove  $AG$  is a superkey using Armstrong's axioms:**

We want to show  $AG \rightarrow ABCDEG$  (all attributes of  $R$ ).

- By reflexivity,  $AG \rightarrow AG$ .
- From  $A \rightarrow BCD$ , by augmentation with  $G$ :

$$AG \rightarrow BCG$$

- From  $BC \rightarrow DE$ , since  $AG \rightarrow BCG$ , we have  $AG \rightarrow BC$ , so by transitivity and augmentation:

$$AG \rightarrow DE$$

- We already have  $AG$  includes  $A, G$ .
- Combine all:

$$AG \rightarrow A, B, C, D, E, G$$

Therefore,  $AG$  functionally determines all attributes in  $R$ , so  $AG$  is a superkey.

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**c. Compute canonical cover for  $F$ :**

Original FDs:

$$F = \{A \rightarrow BCD, \quad BC \rightarrow DE, \quad B \rightarrow D, \quad D \rightarrow A\}$$

**Step 1: Decompose RHS to single attributes**

$$A \rightarrow B, \quad A \rightarrow C, \quad A \rightarrow D$$

$$BC \rightarrow D, \quad BC \rightarrow E$$

$$B \rightarrow D$$

$$D \rightarrow A$$

**Step 2: Remove extraneous attributes on LHS**

Check if  $B$  in  $BC \rightarrow D$  is extraneous:

- Test if  $C \rightarrow D$  using  $F \setminus \{BC \rightarrow D\}$ :

$C^+ = \{C\}$  (no FD starts with  $C$  alone), so no.

Check if  $C$  in  $BC \rightarrow D$  is extraneous:

- Test if  $B \rightarrow D$  exists (it does), so  $C$  is extraneous here.

Remove  $C$  from  $BC \rightarrow D$ :

$$B \rightarrow D$$

This FD already exists, so  $BC \rightarrow D$  is redundant.

Now, the set becomes:

$$A \rightarrow B, \quad A \rightarrow C, \quad A \rightarrow D, \quad BC \rightarrow E, \quad B \rightarrow D, \quad D \rightarrow A$$

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### Step 3: Remove redundant FDs

Check if  $B \rightarrow D$  is redundant.

Try to derive  $B \rightarrow D$  from the others:

-  $B^+ = \{B\}$  without  $B \rightarrow D$ .

No  $B \rightarrow D$  without it, so keep it.

Check if  $A \rightarrow D$  is redundant:

- Try to derive  $A \rightarrow D$  without  $A \rightarrow D$ :

From  $A \rightarrow B$  and  $B \rightarrow D$ , by transitivity:

$$A \rightarrow B \rightarrow D \implies A \rightarrow D$$

So  $A \rightarrow D$  is redundant.

Remove  $A \rightarrow D$ :

Now canonical cover is:

$$A \rightarrow B, \quad A \rightarrow C, \quad BC \rightarrow E, \quad B \rightarrow D, \quad D \rightarrow A$$

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### Final canonical cover:

$$F_c = \{A \rightarrow B, \quad A \rightarrow C, \quad BC \rightarrow E, \quad B \rightarrow D, \quad D \rightarrow A\}$$

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### d. 3NF Decomposition based on canonical cover $F_c$ :

For each FD  $X \rightarrow Y$  in  $F_c$ , create a relation schema  $R_i = X \cup Y$ :

$$R_1 = \{A, B\} \quad \text{from } A \rightarrow B$$

$$R_2 = \{A, C\} \quad \text{from } A \rightarrow C$$

$$R_3 = \{B, C, E\} \quad \text{from } BC \rightarrow E$$

$$R_4 = \{B, D\} \quad \text{from } B \rightarrow D$$

$$R_5 = \{D, A\} \quad \text{from } D \rightarrow A$$

Check if any relation contains a key for  $R$ :

- From earlier,  $AG$  is a key, but  $G$  not present in any above. - So add a relation with a key containing  $AG$  to ensure lossless join and dependency preservation.

Add:

$$R_6 = \{A, G\}$$

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#### e. BCNF Decomposition using original FDs:

Check violations of BCNF: A relation is in BCNF if for every FD  $X \rightarrow Y$ ,  $X$  is a superkey.

- In  $R$ , candidate keys include  $AG$ .

Check FDs on  $R$ :

- $A \rightarrow BCD$ : Is  $A$  a superkey? No, since  $A^+$  from original FDs is  $\{A, B, C, D\}$ , missing  $E$  and  $G$ . So violation.
- $BC \rightarrow DE$ : Is  $BC$  a superkey?  $(BC)^+ = \{B, C, D, E, A\}$  (using  $D \rightarrow A$ ), missing  $G \rightarrow$  No, violation.
- $B \rightarrow D$ :  $B^+$  missing  $E, G$ , no.
- $D \rightarrow A$ :  $D^+$  missing  $E, G$ , no.

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#### BCNF decomposition steps:

1. Decompose using violating FD  $A \rightarrow BCD$ :

$$R_1 = \{A, B, C, D\}, \quad R_2 = \{A, E, G\}$$

2. Check  $R_1$  with FDs projected:

FDs on  $R_1$ :

$$A \rightarrow BCD, \quad B \rightarrow D, \quad D \rightarrow A$$

- $A \rightarrow BCD$  now  $A$  is a key in  $R_1$ , so  $R_1$  in BCNF.

3. Check  $R_2 = \{A, E, G\}$ :

FDs projected:

$$E \rightarrow A$$

- Is  $E$  a key for  $R_2$ ? Check  $E^+$ :

$$E \rightarrow A$$

$E^+ = \{A, E\}$  missing  $G$ , no.

Violation.

Decompose  $R_2$  using  $E \rightarrow A$ :

$$R_{21} = \{E, A\}, \quad R_{22} = \{E, G\}$$

4. Check relations:

- $R_{21}$  with FD  $E \rightarrow A$ ,  $E$  is key  $\rightarrow$  BCNF.
- $R_{22} = \{E, G\}$  has no non-trivial FD  $\rightarrow$  BCNF.

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**Final BCNF relations:**

$$R_1 = \{A, B, C, D\}, \quad R_{21} = \{E, A\}, \quad R_{22} = \{E, G\}$$

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## BCNF Decomposition Example

**Problem:** BCNF Decomposition of Relation  $R$

Consider the schema

$$R = (A, B, C, D, E, G)$$

and the set  $F$  of functional dependencies:

$$F = \{AB \rightarrow CD, \quad B \rightarrow D, \quad DE \rightarrow B, \quad DEG \rightarrow AB, \quad AC \rightarrow DE\}.$$

**Questions:**

1. Explain why the FD  $AB \rightarrow CD$  shows that  $R$  is not in BCNF.
2. Use the BCNF decomposition algorithm starting with  $AB \rightarrow CD$  to decompose  $R$ .
3. Determine whether the resulting decomposition is dependency preserving, and explain your reasoning.

**Solution**

### 1. Why does $AB \rightarrow CD$ show that $R$ is not in BCNF?

A relation schema  $R$  is in BCNF if for every non-trivial FD  $X \rightarrow Y$ ,  $X$  is a superkey of  $R$ .

- The FD  $AB \rightarrow CD$  is non-trivial since  $CD \not\subseteq AB$ . - Check if  $AB$  is a superkey for  $R$ .

To check if  $AB$  is a superkey, compute the closure  $(AB)^+$  under  $F$ :

$$(AB)^+ \supseteq AB$$

Using  $AB \rightarrow CD$ :

$$(AB)^+ = \{A, B, C, D\}$$

Using  $AC \rightarrow DE$  and since  $A, C \in (AB)^+$ :

$$(AB)^+ = \{A, B, C, D, E\}$$

Using  $DE \rightarrow B$ ,  $B$  already included.

$G$  is not in  $(AB)^+$ , so  $AB$  does not determine all attributes in  $R$ .

Hence,  $AB$  is **not** a superkey.

Therefore,  $AB \rightarrow CD$  violates BCNF condition, so  $R$  is **not** in BCNF.

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## 2. BCNF decomposition starting with $AB \rightarrow CD$ :

The BCNF decomposition algorithm splits  $R$  into two relations:

$$R_1 = (ABCD), \quad R_2 = (ABEG)$$

because  $AB \rightarrow CD$  violates BCNF.

### Step 1: Decompose $R$ into

$$R_1 = (A, B, C, D), \quad R_2 = (A, B, E, G)$$

### Step 2: Check if $R_1$ is in BCNF:

Project  $F$  onto  $R_1$ . Relevant FDs with attributes in  $R_1$ :

-  $AB \rightarrow CD$  -  $B \rightarrow D$

Check keys for  $R_1$ :

- Compute  $(AB)^+$  in  $R_1$ :

$$(AB)^+ = \{A, B, C, D\} = R_1$$

So,  $AB$  is a superkey for  $R_1$ .

- For  $B \rightarrow D$ , is  $B$  a superkey for  $R_1$ ?

$$B^+ = \{B, D\}$$

missing  $A, C \rightarrow$  no.

Thus,  $B \rightarrow D$  violates BCNF in  $R_1$ .

### Decompose $R_1$ further on $B \rightarrow D$ :

$$R_{11} = (B, D), \quad R_{12} = (A, B, C)$$

Check if  $R_{11}$  and  $R_{12}$  are in BCNF:

-  $R_{11}$ : FD  $B \rightarrow D$ ,  $B$  is key for  $R_{11}$ , so BCNF.

-  $R_{12}$ : No non-trivial FDs from original  $F$  fully contained in  $R_{12}$ , so BCNF.

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### Step 3: Check $R_2 = (A, B, E, G)$ :

Project FDs onto  $R_2$ :

-  $DE \rightarrow B$  —  $D$  missing in  $R_2$ , so ignore.

-  $DEG \rightarrow AB$  —  $D$  missing, ignore.

-  $AC \rightarrow DE$  —  $C, D$  missing, ignore.

No non-trivial FDs hold fully in  $R_2$ , so  $R_2$  is in BCNF.

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### Final BCNF decomposition:

$$R_{11} = (B, D), \quad R_{12} = (A, B, C), \quad R_2 = (A, B, E, G)$$

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### 3. Dependency preservation:

To check dependency preservation, verify if the union of dependencies from the decomposed relations is equivalent to the original  $F$ .

- Original  $F$  includes  $DE \rightarrow B$  and  $DEG \rightarrow AB$ , which involve attributes spanning multiple decomposed relations and are not fully contained in any single relation.
- These FDs cannot be enforced by checking dependencies locally in each relation. Hence, the decomposition **does not preserve all dependencies**.

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#### Summary:

- $R$  is not in BCNF due to  $AB \rightarrow CD$  where  $AB$  is not a superkey.
- Applying BCNF decomposition starting with  $AB \rightarrow CD$  results in:

$$\{(B, D), (A, B, C), (A, B, E, G)\}$$

- The decomposition is BCNF but not dependency preserving.

## BCNF Decomposition and FD Analysis Example

### Problem Statement

Consider the schema

$$R = (A, B, C, D, E, G)$$

and the set  $F$  of functional dependencies:

$$F = \{A \rightarrow BC, \quad BD \rightarrow E, \quad CD \rightarrow AB\}.$$

Answer the following:

- Find a nontrivial functional dependency containing no extraneous attributes that is logically implied by  $F$ . Explain how you found it.
- Use the BCNF decomposition algorithm to find a BCNF decomposition of  $R$ , starting with  $A \rightarrow BC$ . Explain your steps.
- For your decomposition, state whether it is lossless and explain why.
- For your decomposition, state whether it is dependency preserving and explain why.

### Solution

**(a) Find a nontrivial FD with no extraneous attributes implied by  $F$ :**

Given the functional dependencies:

$$F = \{A \rightarrow BC, \quad BD \rightarrow E, \quad CD \rightarrow AB\}.$$

- Check if  $BD \rightarrow E$  has extraneous attributes:

- $B^+ = \{B\}$ , which does not include  $E$ . So,  $B$  is not extraneous.
- $D^+ = \{D\}$ , which does not include  $E$ . So,  $D$  is not extraneous.

Since  $BD \rightarrow E$  is already given, it is not considered an implied FD.

- Check if  $AD \rightarrow E$  is implied by  $F$ : Compute  $(AD)^+$ :

$$(AD)^+ = \{A, D\} \xrightarrow{A \rightarrow BC} \{A, B, C, D\} \xrightarrow{BD \rightarrow E} \{A, B, C, D, E\}.$$

Thus,

$$AD \rightarrow E$$

is implied by  $F$ .

- Check for extraneous attributes in  $AD \rightarrow E$ :

$$A^+ = \{A, B, C\} \Rightarrow E \notin A^+,$$

so  $D$  is necessary.

$$D^+ = \{D\} \Rightarrow E \notin D^+,$$

so  $A$  is necessary.

**Conclusion:** The FD

$$\boxed{AD \rightarrow E}$$

is a nontrivial, implied functional dependency with no extraneous attributes.

**(b) BCNF decomposition starting with  $A \rightarrow BC$ :**

Since  $A \rightarrow BC$  is a FD and  $A$  alone is **not** a superkey for  $R$  (since  $A^+ = \{A, B, C\}$  missing  $D, E, G$ ), it violates BCNF.

**Decompose  $R$  into:**

$$R_1 = \{A, B, C\} \quad (\text{from } A \rightarrow BC)$$

$$R_2 = \{A, D, E, G\} \quad (\text{the rest})$$

**Step 2: Check BCNF on  $R_1$ :**

-  $R_1$  attributes:  $A, B, C$ . - Relevant FD:  $A \rightarrow BC$  -  $A$  is a superkey for  $R_1$  because  $(A)^+ = \{A, B, C\}$  in  $R_1$ . - So  $R_1$  is in BCNF.

**Step 3: Check BCNF on  $R_2 = (A, D, E, G)$ :**

- Relevant FDs projected on  $R_2$ :

From original  $F$ :

-  $BD \rightarrow E$  — missing  $B$ , not applicable. -  $CD \rightarrow AB$  — missing  $B, C$ , not applicable. -  $A \rightarrow BC$  — missing  $B, C$ , not applicable.

So no nontrivial FDs fully in  $R_2$ .

Hence,  $R_2$  is in BCNF.

**(c) Losslessness of the decomposition:**



The decomposition is lossless if the common attributes between  $R_1$  and  $R_2$  form a key for at least one of the decomposed relations.

$$R_1 \cap R_2 = \{A\}$$

-  $A$  is a key for  $R_1$  (since  $A \rightarrow BC$ ).

Hence, the decomposition is **lossless**.

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**(d) Dependency preservation:**

-  $A \rightarrow BC$  is preserved in  $R_1$ . -  $BD \rightarrow E$  is **not preserved** because  $B$  is in  $R_1$  but  $D, E$  are in  $R_2$ . -  $CD \rightarrow AB$  is **not preserved** for similar reasons.

Thus, the decomposition is **not dependency preserving**.

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**Summary:**

- Nontrivial FD implied:  $AD \rightarrow E$ .
- BCNF decomposition:

$$R_1 = (A, B, C), \quad R_2 = (A, D, E, G)$$

- Decomposition is lossless (common key attribute  $A$ ).
- Decomposition is not dependency preserving.

## 3NF Decomposition Example

Problem: 3NF Decomposition of Relation  $R$

Consider the schema

$$R = (A, B, C, D, E, G)$$

and the set  $F$  of functional dependencies:

$$F = \{AB \rightarrow CD, ADE \rightarrow GDE, B \rightarrow GC, G \rightarrow DE\}.$$

**Tasks:**

- List all candidate keys.
- Compute a canonical cover for  $F$  (show steps).
- Apply the 3NF decomposition algorithm and show the remaining steps.
- Give the final 3NF decomposition and comment on losslessness and dependency preservation.

## Solution

### (a) Candidate keys

We compute closures to find minimal sets that determine all attributes.

- Start with  $B$ :

$$B^+ = \{B\} \xrightarrow{B \rightarrow GC} \{B, G, C\} \xrightarrow{G \rightarrow DE} \{B, G, C, D, E\}.$$

$A$  is missing.

- Try  $AB$ :

$$\begin{aligned} (AB)^+ &= \{A, B\} \xrightarrow{AB \rightarrow CD} \{A, B, C, D\} \\ &\xrightarrow{B \rightarrow GC} \{A, B, C, D, G\} \xrightarrow{G \rightarrow DE} \{A, B, C, D, E, G\} = R \end{aligned}$$

No proper subset of  $\{A, B\}$  closes to  $R$ , so  $\{A, B\}$  is a candidate key.

- Check other small combos:  $A$  alone gives  $\{A, B, C\}$  via  $A \rightarrow ??$  (none), so not a key.  $G$  alone gives  $G, D, E$  only. So no other minimal key appears.

Candidate key:  $\{A, B\}$ .

### (b) Canonical cover for $F$ (step-by-step)

Original FDs:

$$AB \rightarrow CD, \quad ADE \rightarrow GDE, \quad B \rightarrow GC, \quad G \rightarrow DE.$$

#### 1. Split RHS into single attributes

$$AB \rightarrow C, AB \rightarrow D, ADE \rightarrow G, ADE \rightarrow D, ADE \rightarrow E,$$

$$B \rightarrow G, B \rightarrow C, G \rightarrow D, G \rightarrow E$$

**2. Remove trivial FDs:**  $ADE \rightarrow D$  and  $ADE \rightarrow E$  are trivial ( $\text{RHS} \subseteq \text{LHS}$ ), so we drop them. Keep  $ADE \rightarrow G$ .

**3. Remove extraneous attributes on LHS and redundant FDs** -  $AB \rightarrow C$  is redundant because  $B \rightarrow C$  already exists; drop  $AB \rightarrow C$ . - For  $AB \rightarrow D$ : since  $B \rightarrow G$  and  $G \rightarrow D$ , we get  $B \rightarrow D$ . Thus  $A$  is extraneous in  $AB \rightarrow D$  (because  $B \rightarrow D$  already holds), so  $AB \rightarrow D$  is redundant (can be derived). Drop  $AB \rightarrow D$ .

- Now remaining non-trivial, non-redundant FDs:

$$B \rightarrow C, B \rightarrow G, G \rightarrow D, G \rightarrow E, ADE \rightarrow G.$$

#### 4. Combine FDs with same LHS

$$B \rightarrow CG, \quad G \rightarrow DE, \quad ADE \rightarrow G.$$

This set has no extraneous LHS attributes (quick checks: neither attribute of  $ADE$  can be removed while still deriving  $G$ ), so it is a canonical cover.

$$F_c = \{B \rightarrow CG, G \rightarrow DE, ADE \rightarrow G\}.$$

### (c) 3NF decomposition algorithm (steps)

Algorithm: For each FD  $X \rightarrow Y$  in the canonical cover, create relation  $R_X = X \cup Y$ . Ensure some relation contains a candidate key.

From  $F_c$ :

- From  $B \rightarrow CG$  create  $R_1 = (B, C, G)$ . - From  $G \rightarrow DE$  create  $R_2 = (G, D, E)$ . -

From  $ADE \rightarrow G$  create  $R_3 = (A, D, E, G)$ .

Check for candidate key  $\{A, B\}$ : none of  $R_1, R_2, R_3$  contains both  $A$  and  $B$ . So add relation holding a key:

- Add  $R_4 = (A, B)$ .

Optional optimization: if any relation is a subset of another, we may drop the subset. Here  $R_2 = (G, D, E)$  is a subset of  $R_3 = (A, D, E, G)$ . We may drop  $R_2$  and keep  $R_3$  instead.

#### (d) Final decomposition and comments

A compact, dependency-preserving, lossless 3NF decomposition is:

$$R_1(B, C, G), \quad R_2(A, D, E, G), \quad R_3(A, B).$$

(If you prefer the unoptimized listing, use  $(B, C, G)$ ,  $(G, D, E)$ ,  $(A, D, E, G)$ ,  $(A, B)$ .)

**Losslessness:** We included relation  $R_3(A, B)$  that contains a candidate key  $\{A, B\}$ . Therefore the decomposition is *lossless*.

**Dependency preservation:** Each FD from the canonical cover appears in some relation:

- $B \rightarrow CG$  is preserved in  $R_1(B, C, G)$ .
- $G \rightarrow DE$  is preserved in  $R_2(A, D, E, G)$  (or in  $R_2(G, D, E)$  if you kept it).
- $ADE \rightarrow G$  is preserved in  $R_2(A, D, E, G)$ .

Since the canonical-cover FDs are preserved, the original set  $F$  is preserved as well. Thus the decomposition is *dependency preserving*.

□

## 3NF Decomposition Example

Problem: 3NF Decomposition of Relation  $R$

Consider the schema

$$R = (A, B, C, D, E, G, H)$$

and the set  $F$  of functional dependencies:

$$F = \{AB \rightarrow CD, D \rightarrow C, DE \rightarrow B, DEH \rightarrow AB, AC \rightarrow DC\}.$$

**Tasks:**

- List all candidate keys.
- Compute a canonical cover for  $F$  (show steps).

- (c) Apply the 3NF decomposition algorithm and explain.
- (d) Give the final 3NF decomposition, and comment on losslessness and dependency preservation.

### Solution

#### (a) Candidate Keys

We compute closures:

1.  $(DEH)^+ = \{D, E, H\} \xrightarrow{DE \rightarrow B} \{B, D, E, H\} \xrightarrow{DEH \rightarrow AB} \{A, B, D, E, H\} \xrightarrow{AB \rightarrow CD} \{A, B, C, D, E, H\} \xrightarrow{\text{add } G? \text{ none}} \text{Missing } G, \text{ so not a key.}$
2. Try  $DEHG$ :

$$(DEHG)^+ \supseteq DEH \cup \{G\} \Rightarrow \{A, B, C, D, E, G, H\} = R$$

So  $DEHG$  is a key.

No proper subset of  $DEHG$  gives all attributes, hence:

Candidate Key:  $\{D, E, G, H\}$

#### (b) Canonical Cover

**Step 1: Split RHS to single attributes**

$$\begin{aligned} AB &\rightarrow C, & AB &\rightarrow D, \\ D &\rightarrow C, \\ DE &\rightarrow B, \\ DEH &\rightarrow A, & DEH &\rightarrow B, \\ AC &\rightarrow D, & AC &\rightarrow C \end{aligned}$$

**Step 2: Remove extraneous attributes from LHS**

- In  $AB \rightarrow C$ : Check if  $A \rightarrow C$  or  $B \rightarrow C$  holds — neither does, so keep  $AB$ . - In  $DEH \rightarrow A$ : Test  $DE \rightarrow A$  — does not hold, so keep  $DEH$ . - Similarly check for others.

**Step 3: Remove redundant dependencies**

-  $AB \rightarrow C$  is redundant since  $AB \rightarrow D$  and  $D \rightarrow C$  already imply  $AB \rightarrow C$ . -  $AC \rightarrow C$  is trivial, remove it.

**Canonical Cover:**

$$F_c = \{AB \rightarrow D, D \rightarrow C, DE \rightarrow B, DEH \rightarrow A, AC \rightarrow D\}$$

#### (c) Apply 3NF Decomposition Algorithm

Step 1: For each FD in  $F_c$ , create a relation with LHS  $\cup$  RHS.

$$\begin{array}{ll}
R_1 = (A, B, D) & \text{from } AB \rightarrow D \\
R_2 = (D, C) & \text{from } D \rightarrow C \\
R_3 = (D, E, B) & \text{from } DE \rightarrow B \\
R_4 = (D, E, H, A) & \text{from } DEH \rightarrow A \\
R_5 = (A, C, D) & \text{from } AC \rightarrow D
\end{array}$$

Step 2: Ensure at least one relation contains a key.

The key  $\{D, E, G, H\}$  is not in any existing relation, so we add:

$$R_6 = (D, E, G, H)$$

Step 3: Remove redundant relations (those contained in others).

After removing subsumed ones, we get:

$$R_1 = (A, B, D), R_2 = (D, C), R_4 = (D, E, H, A), R_6 = (D, E, G, H)$$

#### (d) Final 3NF Decomposition

$$\{(A, B, D), (D, C), (D, E, H, A), (D, E, G, H)\}$$

**Losslessness:** Decomposition is lossless because for each decomposition step, the common attribute set is a key in one of the decomposed relations.

**Dependency Preservation:** All FDs in  $F_c$  are preserved in at least one relation, so the decomposition is dependency-preserving.

## BCNF

### BCNF but **Not** Lossless

**Schema and FD:**

$$R(A, B, C), \quad F = \{A \rightarrow B\}.$$

**Decomposition:**

$$R_1(A, B), \quad R_2(B, C).$$

**Each component is BCNF:**

- In  $R_1$ :  $A \rightarrow B$  holds and  $A$  is a key of  $R_1 \Rightarrow R_1$  is BCNF.
- In  $R_2$ : no non-trivial FD with LHS  $\subseteq \{B, C\}$  from  $F \Rightarrow R_2$  is BCNF.

**But decomposition is *not* lossless:** intersection  $R_1 \cap R_2 = \{B\}$ . Lossless test requires  $\{B\} \rightarrow R_1$  or  $\{B\} \rightarrow R_2$ ; neither holds (no  $B \rightarrow A$ , no  $B \rightarrow C$  in  $F$ ). Thus not guaranteed lossless.

**Concrete counterexample (shows spurious tuples):** take an instance  $r$  of  $R$  that satisfies  $A \rightarrow B$ :

$$r = \{(A = 1, B = 2, C = 3), (A = 4, B = 2, C = 5)\}.$$

Projections:

$$\pi_{A,B}(r) = \{(1, 2), (4, 2)\}, \quad \pi_{B,C}(r) = \{(2, 3), (2, 5)\}.$$

Join:

$$\pi_{A,B}(r) \bowtie \pi_{B,C}(r) = \{(1, 2, 3), (1, 2, 5), (4, 2, 3), (4, 2, 5)\},$$

which contains spurious tuples  $(1, 2, 5), (4, 2, 3)$ . Hence the decomposition is lossy.

### BCNF and Lossless Example

**Schema and FDs:**

$$R(A, B, C), \quad F = \{A \rightarrow B, B \rightarrow C\}.$$

**Decomposition:**

$$R_1(A, B), \quad R_2(A, C).$$

**Verify BCNF:**

- In  $R_1(A, B)$ :  $A \rightarrow B$  holds, and  $A$  is a key for  $R_1 \Rightarrow R_1$  is BCNF.
- In  $R_2(A, C)$ : since  $A \rightarrow B$  and  $B \rightarrow C$  imply  $A \rightarrow C$ , we have  $A \rightarrow C$  in  $R_2$ .  $A$  is a key for  $R_2 \Rightarrow R_2$  is BCNF.

**Losslessness:** intersection  $R_1 \cap R_2 = \{A\}$ . Since  $A \rightarrow B$  (in  $R_1$ ) and  $A \rightarrow C$  (in  $R_2$ ), we have  $\{A\}$  is a key for both  $R_1$  and  $R_2$  (or at least determines all attributes of each component). Thus the join is lossless.

**Dependency preservation:** all original FDs ( $A \rightarrow B, B \rightarrow C$ ) are preserved:

- $A \rightarrow B$  in  $R_1$ .
- $B \rightarrow C$  can be checked via the join: but note canonical-cover viewpoint gives  $A \rightarrow C$  in  $R_2$ , and together the dependencies are preserved (or equivalently the canonical cover  $\{A \rightarrow B, A \rightarrow C\}$  is preserved).

**Concrete example (shows lossless behavior):** take

$$r = \{(1, 2, 3)\}$$

with  $A = 1, B = 2, C = 3$ . Projections:

$$\pi_{A,B}(r) = \{(1, 2)\}, \quad \pi_{A,C}(r) = \{(1, 3)\}.$$

Join  $\{(1, 2)\} \bowtie \{(1, 3)\} = \{(1, 2, 3)\}$  — no spurious tuples.

### Every 2-attribute schema is in BCNF

**Claim.** Let  $R$  be a relation schema with exactly two attributes, say  $R = (A, B)$ . Then  $R$  is in BCNF for *any* set  $F$  of functional dependencies.

**Proof.** By definition,  $R$  is in BCNF iff for every nontrivial FD  $X \rightarrow Y$  implied by  $F$  (i.e.  $F \models X \rightarrow Y$  and  $Y \not\subseteq X$ ), the left-hand side  $X$  is a superkey of  $R$ . Since  $R$  has only two attributes, the only possible (non-empty) subsets of attributes are  $\{A\}$ ,  $\{B\}$ , and  $\{A, B\}$ . Consider any nontrivial FD  $X \rightarrow Y$ .

- If  $X = \{A, B\} = R$ , then  $X$  trivially determines all attributes of  $R$ , so  $X$  is a superkey.
- If  $X = \{A\}$  and  $X \rightarrow Y$  is nontrivial, then  $Y$  must contain the other attribute  $B$  (because the only attributes are  $A$  and  $B$ , and nontrivial means  $B \notin X$  initially). Hence  $A \rightarrow B$  holds, so  $A^+ = \{A, B\} = R$ ; thus  $A$  is a key (superkey).
- Similarly, if  $X = \{B\}$  and  $X \rightarrow Y$  is nontrivial then  $B \rightarrow A$  holds, so  $B$  is a key.

In every case a nontrivial determinant  $X$  is a superkey. Therefore all nontrivial FDs satisfy the BCNF requirement and  $R$  is in BCNF.  $\square$

**Remark.** Intuitively, with only two attributes any nontrivial dependency between them makes the left side determine the whole relation, so the determinant must be a key.